but also, and even more importantly, we haven’t said anything about how we are going to split a rectangle at points 2 and 3.

As a matter of fact, this choice, together with the heuristic for choosing the insertion subtree, determine the behavior and shape (not to mention performance) of the R-tree.

Several heuristics have been studied over the years, each one aiming to optimize one or more usage of the tree; the split heuristics can be particularly complicated for internal nodes, because we don’t just partition points, but k-dimensional shapes: figure 10.7 shows how easily a naïve choice could lead to inefficient splits.

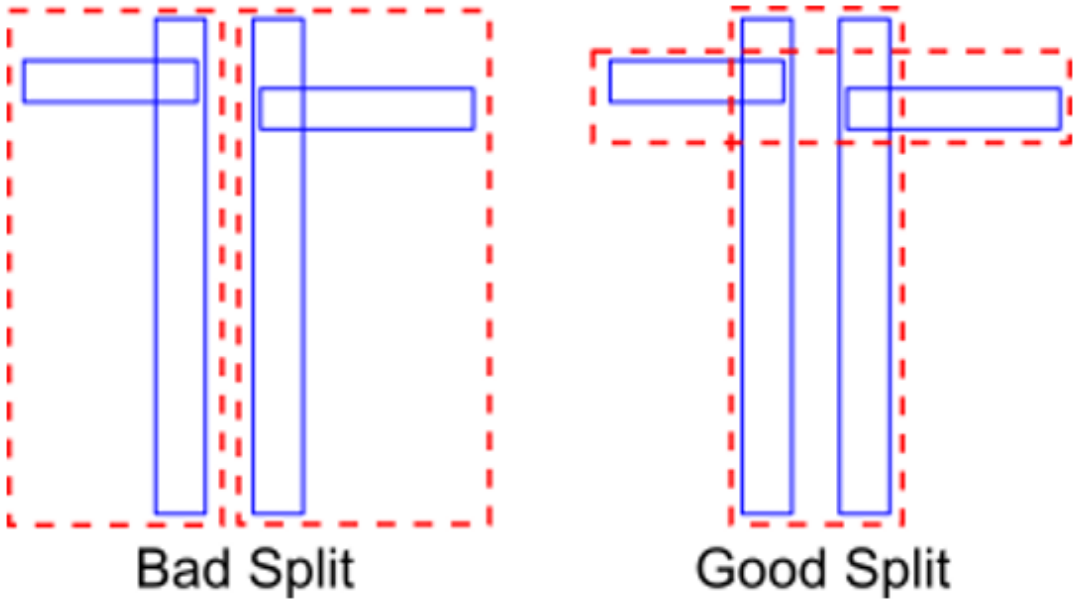


Figure 10.7 An example of bad and good splits of an internal node’s rectangle, taken from the original paper by Guttman.